

ORIGINAL ARTICLE

Sun exposure and protection habits: Self-reported attitudes, knowledge and behaviours

Henry W. Lim¹ | Marketa Saint Aroman²  | Charbel Skayem^{3,4}  | Bruno Halioua⁵  |
 Nuria Perez Cullell² | Yaron Ben Hayoun⁶ | Catherine Baissac⁷ | Christina Bergqvist⁸  |
 Charles Taieb⁹  | Marie-Aleth Richard¹⁰  | Khaled Ezzedine^{8,11} 

¹Department of Dermatology, Henry Ford Health, Detroit, Michigan, USA

²Dermocosmetics Care & Personal Care, Pierre Fabre, Paris, France

³Faculty of Medicine, Sorbonne University, Paris, France

⁴Hôpitaux de Paris (AP-HP), Paris Saclay University, Ambroise Paré Hospital, Boulogne Billancourt, France

⁵Private Practice, Paris, France

⁶European Market Maintenance Assessment, Tel Aviv, Israel

⁷Patient Centricity Department, Pharma, Dermocosmetics Care & Personal Care, Pierre Fabre, Paris, France

⁸Department of Dermatology, AP-HP, Henri Mondor University Hospital, UPEC, Créteil, France

⁹Patient Priority Department, European Market Maintenance Assessment, Fontenay-sous-Bois, France

¹⁰Dermatology Department, CEReSS-EA 3279, Research Centre in Health Services and Quality of Life Aix Marseille University, University Hospital Timone, Assistance Publique Hôpitaux de Marseille, APHM, Marseille, France

¹¹EA 7379 EpiderME, Université Paris-Est Créteil (UPEC), Créteil, France

Correspondence

Charles Taieb, European Market Maintenance Assessment, 94120 Fontenay-sous-Bois, France.

Email: charles.taieb@emma.clinic

Funding information

Patient Centricity by Pierre Fabre

Abstract

Background: As exposure to UV radiation is the primary modifiable environmental risk factor associated with skin cancer, it remains the principal focus of most prevention strategies. Numerous sun protection campaigns have been implemented worldwide; however, their impact on the actual incidence and mortality rates of skin cancer seems to be limited. To create successful skin cancer prevention campaigns, it is important to have a comprehensive understanding of individuals' attitudes and behaviours regarding sun protection. The aim of the current study was to determine and report on the prevalence of self-reported attitudes, knowledge and behaviours regarding two of the major sun protection recommendations—avoidance of sun exposure and use of sunscreens—in an international representative sample across five continents.

Methods: This cross-sectional study was conducted in 20 countries using a web-based online survey.

Findings: A total of 50,552 individuals, comprising 25,388 men (50.22%) and 25,164 women (49.78%), participated in the survey. Among them, 83.2% reported having been voluntarily exposed to the sun (for sun-basking reasons) at least once in the last 12 months, and 47.96% acknowledged being exposed to the sun between the hours of 10 AM and 4 PM. The primary reason for non-adherence was that these hours were the most convenient times (32.28%). Only 24.05% reported applying sunscreen every 2 h when outdoors. Forgetfulness was the primary reason as provided by 27.79% of participants. Males and older age groups were less likely to adopt sun-protective behaviours around the world. Forgetfulness and the challenges posed by time constraints seem to be the biggest barriers to proper adherence.

Interpretation: These findings should prompt the collaboration with health authorities and the manufacturers to enhance adherence by setting reasonable sunscreen prices and creating formulations that make their application less burdensome.

INTRODUCTION

Skin cancer is a growing global concern, with an incidence and prevalence increasing at an alarming rate across the

world.^{1–3} According to the World Health Organization (WHO), one in every three diagnosed cancers is a skin cancer, and between 2 and 3 million keratinocyte cancers (KC) and 132,000 melanoma skin cancers are diagnosed

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial](https://creativecommons.org/licenses/by-nc/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

© 2024 The Authors. *Journal of the European Academy of Dermatology and Venereology* published by John Wiley & Sons Ltd on behalf of European Academy of Dermatology and Venereology.

globally each year.⁴ One of the key factors contributing to the surge in skin cancer is excessive exposure to ultraviolet (UV) radiation.⁵ According to the WHO, the rise in outdoor activities and recreational exposure has played a significant role in the increased incidence of skin cancers in recent decades. Approximately 95% of all skin melanoma cases and deaths in the United States are attributable to UV radiation.⁶ Similarly, it has been estimated in several countries that melanomas are among the cancer types with the highest proportion of preventable cases, as UV damage is mostly avoidable.^{7,8}

As exposure to UV radiation is the primary modifiable environmental risk factor associated with skin cancer, it remains the principal focus of most prevention strategies implemented worldwide. Different agencies, including the WHO, have provided evidence-based guidelines for sun protection.⁹ These include avoiding sun exposure when UV rays are the strongest (between 10 AM and 4 PM); seeking shade; wearing protective clothing, hat and sunglasses; using broad-spectrum sunscreens every 2 h when outdoors; and avoiding sunlamps and indoor tanning. Numerous sun protection campaigns have been implemented worldwide, and have increasingly utilized digital platforms due to their potential wide reach and cost effectiveness.^{10–12} While these programs have shown success in reducing the occurrence of sunburns, their impact on the actual incidence and mortality rates of skin cancer seems to be limited.^{13,14}

As the incidence of skin cancer continues to rise globally, it is crucial to intensify efforts aimed at promoting and encouraging individuals to adopt protective behaviours that reduce their risk of skin cancer. To create successful skin cancer prevention campaigns, it is important to have a comprehensive understanding of individuals' attitudes and behaviours regarding sun protection. By elucidating the reasons behind non-compliance, we can develop tailored interventions that address the specific barriers individuals face, with the ultimate aim of promoting widespread adoption of sun protective measures use and reducing the burden of UV radiation-related diseases.

The aim of the current study was to determine and report on the prevalence of self-reported attitudes, knowledge and behaviours regarding two of the major sun protection recommendations—avoidance of sun exposure and use of sunscreens—in an international representative sample across five continents.

METHODS

Study population

This multinational, cross-sectional study was conducted in 20 countries using a web-based online survey by a polling company (HC Conseil Paris, France) between January and April 2023. Institutional review board approvals were not required since the study did not involve any clinical examination and used anonymized data.

For each of the 20 countries (Table 1), the study recruited a representative sample of the general population over 16 years of age. The selection of participants aimed to ensure national representativeness by considering factors such as age, gender, administrative region, environment (including large cities, towns and rural areas) and income distribution in each country, as previously described in previous studies.^{15,16} The survey covered the population of 20 countries representing more than 50% of the world's inhabitants. Specifically focusing on individuals aged 16 years or above, the total population from which the sample was derived amounted to 3,213,401,122.

Information regarding the sociodemographic composition of the population in each country was obtained from the Eurostat database, which was updated to reflect the most recent data available during the survey in 2020. Additionally, supplementary data from the United Nations, Department of Economic and Social Affairs, Population Division were also incorporated. World Population Prospects: the 2019 Revision. <https://population.un.org/wpp/>.

Data were collected via the Internet from a randomly chosen subset of 50,552 individuals from the pool of 900,000 Internet users aged 16 and above and willing to take part in the surveys. Each participant was initially contacted via email, and if contact could not be established, another individual with similar characteristics was randomly selected as a potential replacement.

Questionnaire

The participants were requested to fill out a comprehensive digital questionnaire that consisted of a structured set of questions covering a range of sociodemographic aspects, such as gender, age and other relevant factors. In countries where it was legally permissible, an inquiry regarding ethnic origin was included. Participants were asked about voluntary sun exposure for sun-basking reasons in the past 12 months, sun exposure between 10 AM and 4 PM, and the use of broad-spectrum sunscreens. Subsequently, each participant was queried about the reasons behind their adherence or non-adherence to both sets of recommendations.

Statistical analysis

The 20 countries were grouped into five regions: North America and Australia (Australia, Canada and United States), Latin America (Brazil and Mexico), Europe (Denmark, France, Germany, Italy, Israël, Poland, Portugal and Spain), Africa (Kenya, Senegal, South Africa and United Arab Emirates) and Asia (China, India and South Korea). These five regions were defined and consolidated during a meeting involving three of the authors (HL, CBa and CT.).

In each country, the total population of individuals, as well as the total population by sex and age, were calculated

TABLE 1 Total patient population.

	Country	Total number of inhabitants (in 2021)	Number of inhabitants 16 years and older (in 2021)	Number of men 16 years and older (in 2021)	Number of women 16 years and older (in 2021)	Proposed sample sizes per country
North America and Australia	Australia	26,439,103	21,372,242	10,528,875	10,843,367	2000
	Canada	38,781,280	32,389,906	16,005,710	16,384,196	2500
	USA	334,319,658	267,302,837	131,408,588	135,894,248	5000
Latin America	Brazil	216,422,436	170,025,851	82,582,233	87,443,618	4001
	Mexico	128,455,559	95,440,511	45,893,170	49,547,341	2500
Asia	China	1,425,671,344	1,169,958,264	590,459,533	579,498,731	5000
	India	1,428,627,653	1,047,596,423	538,029,159	509,567,265	3000
	South Korea	51,784,047	45,509,238	22,616,052	22,893,186	2500
Europe and Israël	Denmark	5,910,901	4,899,008	2,421,671	2,477,338	1000
	France	64,756,570	52,936,023	16,005,710	16,384,196	4000
	Germany	83,294,620	70,881,658	34,733,106	36,148,552	4000
	Israel	9,174,512	6,467,227	3,190,057	3,277,170	1000
	Italy	58,870,750	51,118,409	24,738,447	26,379,963	4001
	Poland	41,026,055	34,458,521	16,512,116	17,946,405	2500
	Portugal	10,247,592	8,809,552	4,101,985	4,707,567	1000
	Spain	47,519,618	40,540,987	19,699,164	20,841,823	4000
Africa	Kenya	55,100,574	33,325,569	16,374,936	16,950,633	500
	Senegal	17,763,154	10,071,911	4,942,551	5,359,981	300
	South Africa	60,414,485	42,300,029	20,234,950	22,065,079	1000
	UAE	9,516,860	7,996,954	5,799,524	2,197,430	750

by direct extrapolation of the proportionally stratified sample proportions in each country.

Descriptive characteristics of the study population were conducted, and qualitative variables were reported as the number (percentage) whereas quantitative variables were reported as the median and/or the mean SD. Participants were grouped into three age categories: 16–34 years, 35–54 years and ≥ 55 years. Sun protection behaviours and demographic characteristics were compared using the chi-squared test or the Fisher's exact test as needed.

All analyses were conducted in RStudio using R 4.1.3.

RESULTS

Among the participants included in the survey, there were a total of 50,552 individuals, comprising 25,388 males (50.22%) and 25,164 females (49.78%).

Prevalence of sun exposure

Among study participants, $83.2\% \pm 0.33\%$ [81.97%, 82.63%] reported having been voluntarily exposed to the sun (for sun-basking reasons) at least once in the last 12 months, with higher rates in males ($85.04\% \pm 0.44\%$ [84.60%,

85.48%]) compared with females ($82.38\% \pm 0.47\%$ [81.91%, 82.85%]) ($p < 0.001$). This prevalence was highest in Africa ($90.97\% \pm 1.11\%$ [89.86%, 92.08%]), followed by Latin America ($88.07\% \pm 0.78\%$ [87.29%, 88.85%]), Asia ($85.03\% \pm 0.68\%$ [84.35%, 85.71%]), North America/Australia ($78.02\% \pm 0.83\%$ [77.19%, 78.85%]) and Europe ($71.62\% \pm 0.60\%$ [71.02%, 72.22%]).

The prevalence of sun exposure was highest in the 16- to 34-year group (87.51%), with a gradual decrease in prevalence as age increased (82.41% and 77.51% in the 35- to 54-year and ≥ 55 years age groups, respectively) ($p < 0.001$). The higher prevalence of sun exposure among males compared with females and the trend of higher sun exposure among the younger age groups were consistently observed across all five regions (Figure 1).

A total of 47.96% of the participants acknowledged being exposed to the sun between the hours of 10 AM and 4 PM, with higher rates in males (49.33%) compared with females (46.53%) ($p < 0.001$). This prevalence was highest in Africa (60.68%), followed by North America/Australia (57.10%), Asia (46.71%), Latin America (46.55%) and Europe (45.53%). The highest prevalence of sun exposure during these hours was observed in the 16- to 34-year age group (53.75%), with a gradual decrease in prevalence as age increased (45.08% and 41.18% in the 35- to 54-year and ≥ 55 years age groups, respectively) ($p < 0.001$). Once again, the higher prevalence

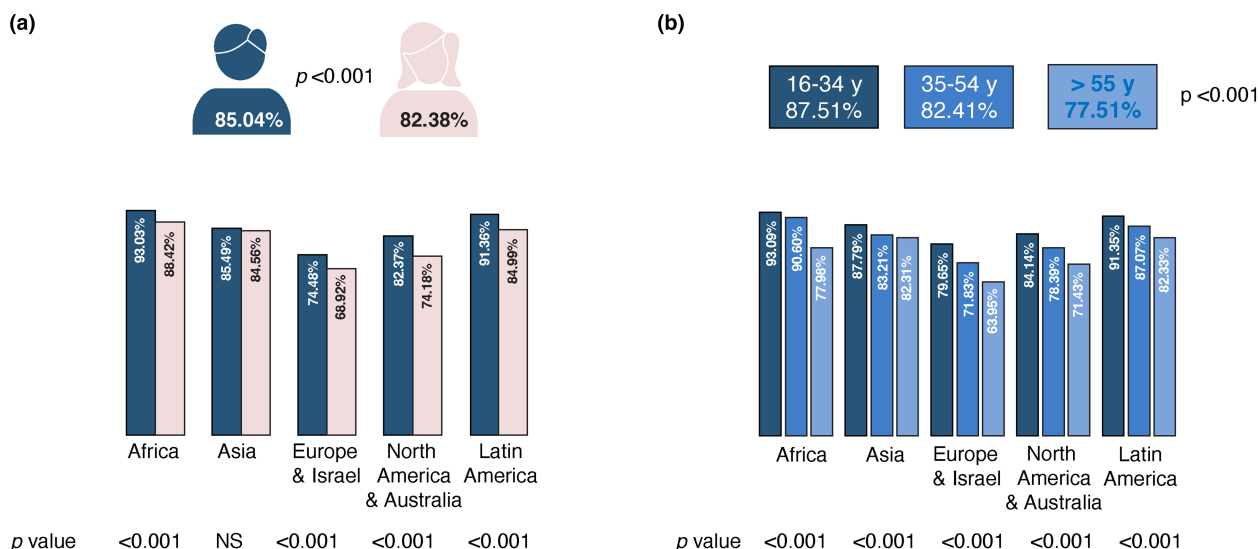


FIGURE 1 Prevalence of sun exposure in the last 12 months according to (a) sex and (b) age.

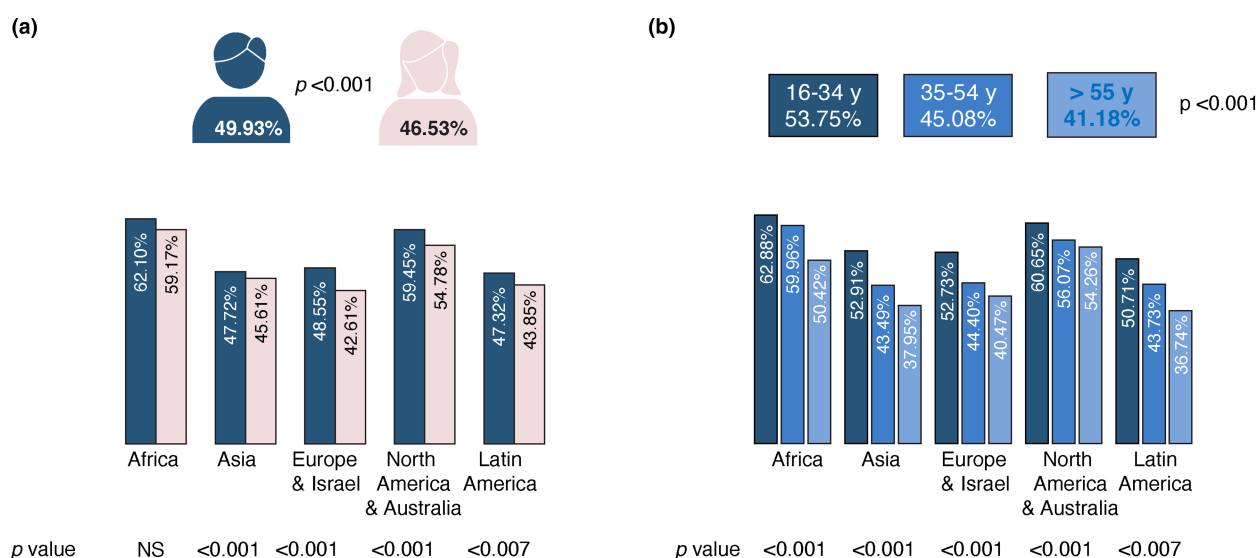


FIGURE 2 Prevalence of sun exposure in the last 12 months between 10 AM and 4 PM according to (a) sex and (b) age.

of sun exposure between 10 AM and 4 PM among males and the younger age groups was consistently observed across all five regions (Figure 2).

Underlying reasons contributing to the failure to avoid sun exposure between 10 AM and 4 PM

The primary reason was that these hours corresponded to the most convenient times (32.28%), followed by the fact that these are the hours when they are available (25.68%) and these are the most pleasant hours (17.88%). Additional reasons included a lack of belief in the prevention messages (5.24%), a lack of comprehension of the prevention message (4.62%) and an uncertainty regarding their own motivations for engaging in such behaviour (2.21%).

The distribution of responses to this question based on sex is detailed in Figure 3.

Prevalence of sunscreen application every 2 h when outdoors

Out of the study participants, only 24.05% reported applying sunscreen every 2 h when outdoors. This prevalence was highest in the Latin America regions (29.98%), followed by Asia (25.08%), Europe (22.13%), North America/Australia (16.48%) and Africa (13.93%). Females demonstrated a significantly higher prevalence (28.28%) of this behaviour compared to males (19.87%) ($p < 0.001$). The higher prevalence in female participants was observed across all five regions (Figure 4). The prevalence of applying sunscreen every 2 h was highest in

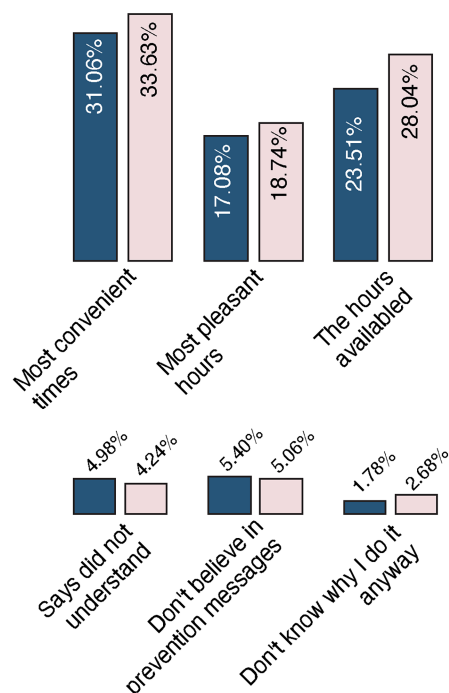


FIGURE 3 Underlying reasons contributing to the failure to avoid sun exposure between 10 AM and 4 PM according to sex.

the 16- to 34-year age group (27.04%) and exhibited a gradual decline with increasing age (25.88% and 12.67% in the 35- to 54-year and ≥ 55 years age groups, respectively) ($p < 0.001$).

Underlying reasons contributing to the failure of applying sunscreen every 2 h when outdoors

Participants who indicated not applying sunscreen every 2 h when outdoors (75.95%) were queried about the reasons behind their behaviour. As reported by 27.79% of participants, the primary reason provided was that they simply did not think of applying sunscreen every 2 h ('it does not occur to me').

Additional reasons mentioned included finding the task too tedious (18.59%), perceiving sunscreen products as too costly (17.63%), considering them ineffective (16.64%) and lacking awareness of this specific recommendation (15.13%). The distribution of responses to this question based on regions and sex is detailed in the Figure 5.

There was a higher percentage of males compared to females who reported: not thinking of applying the sunscreen every 2 h (28.41% vs. 26.87%), not finding it useful (18.27% vs. 14.20%) and not being aware of this recommendation (16.43% vs. 10.48%). A higher percentage of females (18.93%) perceived sunscreen products as too costly as compared to males (16.54%). Africa exhibited the highest percentage of participants reporting that sunscreen was too expensive (34.40%), and not being aware of this recommendation (23.39%).

Underlying reasons contributing to the practice of sun protection

The prevailing incentive was to avoid sunburn (53.38%), followed by the desire to protect against accelerated skin ageing (40.21%), skin tumours prevention (27.56%), spending more time in the sun (21.19%) and considerations based on a history of sunburn (17.51%). The distribution of responses to this question based on age, sex and region is detailed in the Figure 6.

Protection against accelerated skin ageing was reported by a higher percentage of females (44.13%) compared to males (35.77%) and was mostly a concern of the 35- to 54-year age group (41.52% vs. 40.27% and 37.07% in 16- to 34-year and ≥ 55 years age groups, respectively, $p < 0.001$).

DISCUSSION

To our knowledge, this is the first and largest population-based study examining the prevalence of self-reported attitudes and behaviours regarding two of the major sun protection recommendations in an international representative sample across five continents. The study we are reporting on herein covers a representative sample of 20 countries, representing 50% of the world's inhabitants and leading to a total of 50,552 participants aged ≥ 16 years.

Almost half of the participants (48%) acknowledged being voluntarily exposed to the sun between the hours of 10 AM and 4 PM. Among these individuals, almost a third reported that these were the most convenient times (32%), and a quarter reported that these are the hours when they are the most available (26%). While the timing of lunch breaks can vary depending on cultural norms, workplace policies and individual schedules, in many countries, they typically occur around midday, between 12:00 PM and 1:00 PM. One potential solution could involve structural strategies such as implementing scheduling processes that assign outdoor events to periods of the day when UV radiation is minimal or providing shade or indoor alternatives for such events.

Strong evidence suggests that regular sunscreen use decreases the risks of melanoma and keratinocyte carcinoma.^{17,18} The overall consensus is to apply a broad-spectrum sunscreen 15–20 min before exposure, and reapply every 2 h when outdoors.¹⁹ Yet in our study, more than three quarters (76%) of participants did not adhere to the recommended practice of applying sunscreen every 2 h. More than a quarter of these individuals cited forgetfulness as the primary reason for not applying sunscreen. This highlights the need to enhance sun protection campaigns to improve people's overall recall. A randomized controlled trial (RCT) was conducted back in 2007 on 70 volunteers from the general community in Boston USA to evaluate whether using text-message reminders would improve sunscreen use. At the end of the 6 weeks study period, the 35 participants who did not receive reminders had a mean daily adherence rate

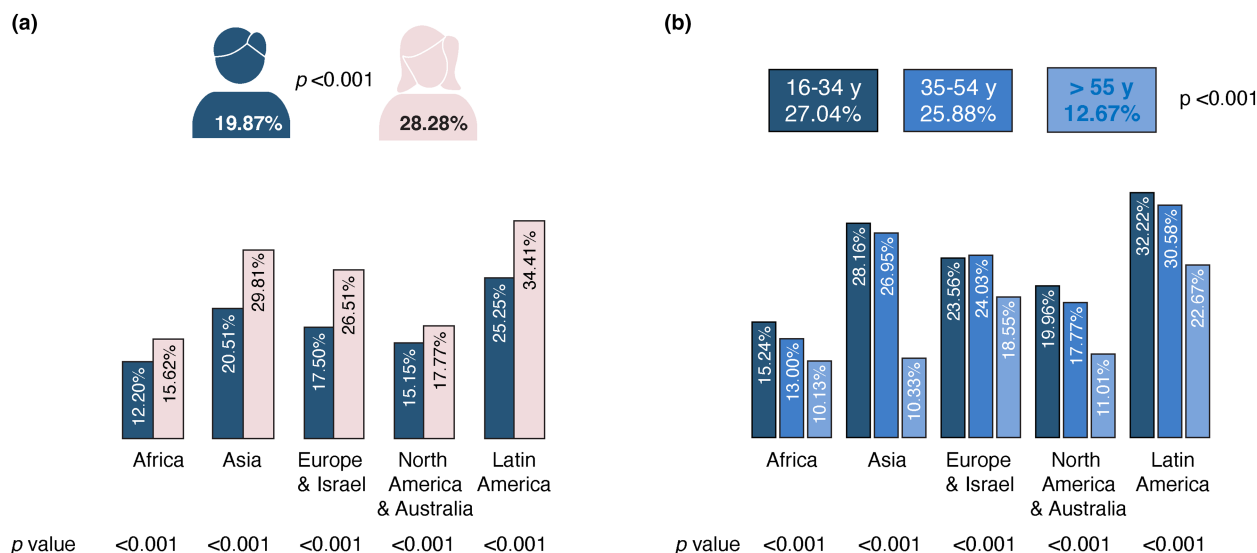


FIGURE 4 Prevalence of sunscreen application every 2 h according to (a) sex and (b) age.

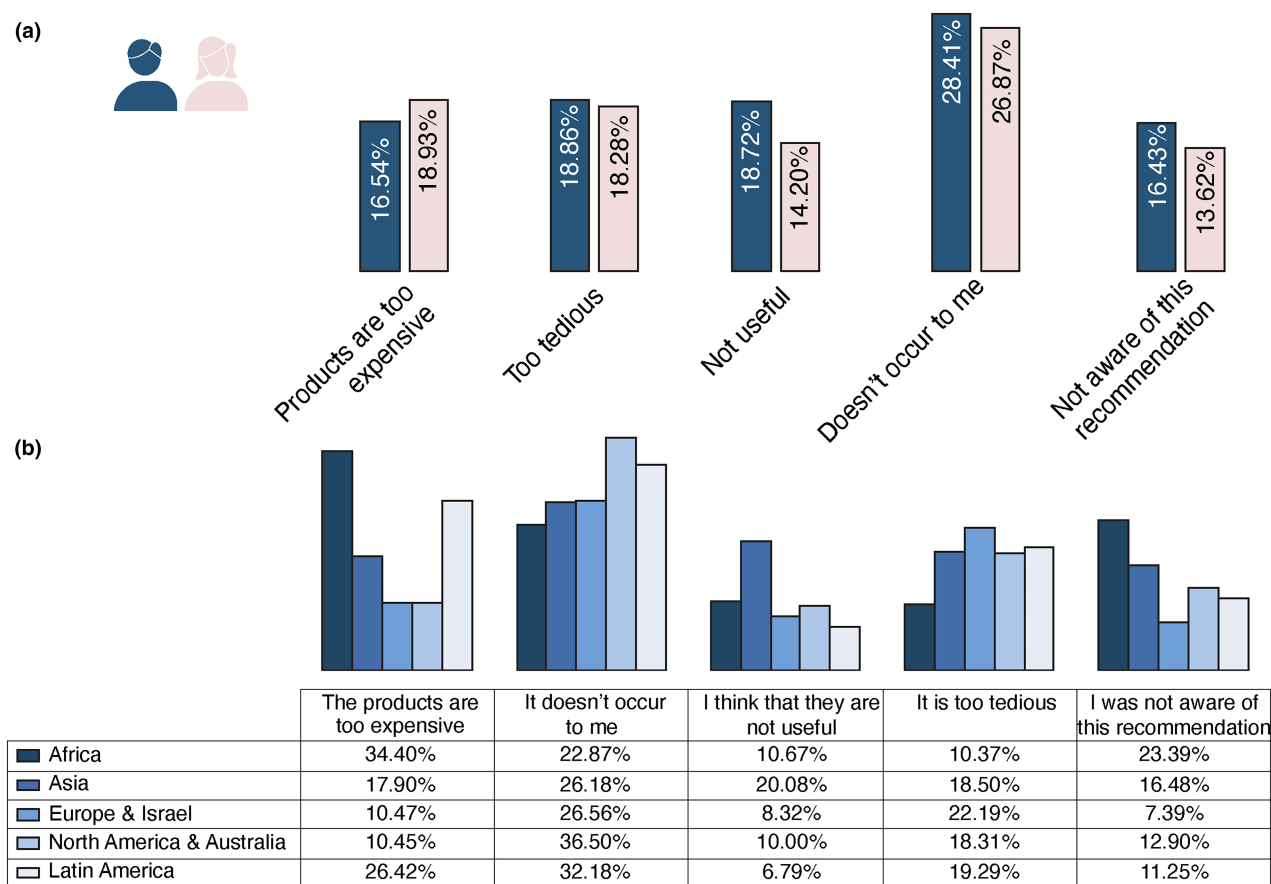


FIGURE 5 Underlying reasons contributing to the failure of applying sunscreen every 2 h when outdoors according to (a) sex and (b) region.

of 30% whereas the 35 participants who received daily text messages reminders had a mean daily adherence rate of 56% ($p < 0.001$).²⁰ A systematic review on 5 RCTs that assessed effects of SMS text reminders; however, did not find a significant benefit on sunburn, sunscreen use and sun protection

habits.²¹ Nevertheless, utilizing platforms like social media nowadays could offer a potentially more effective approach to reach a wider audience and increase awareness about the importance of sunscreen application.²² Indeed, social media platforms provide more sophisticated tools for targeted

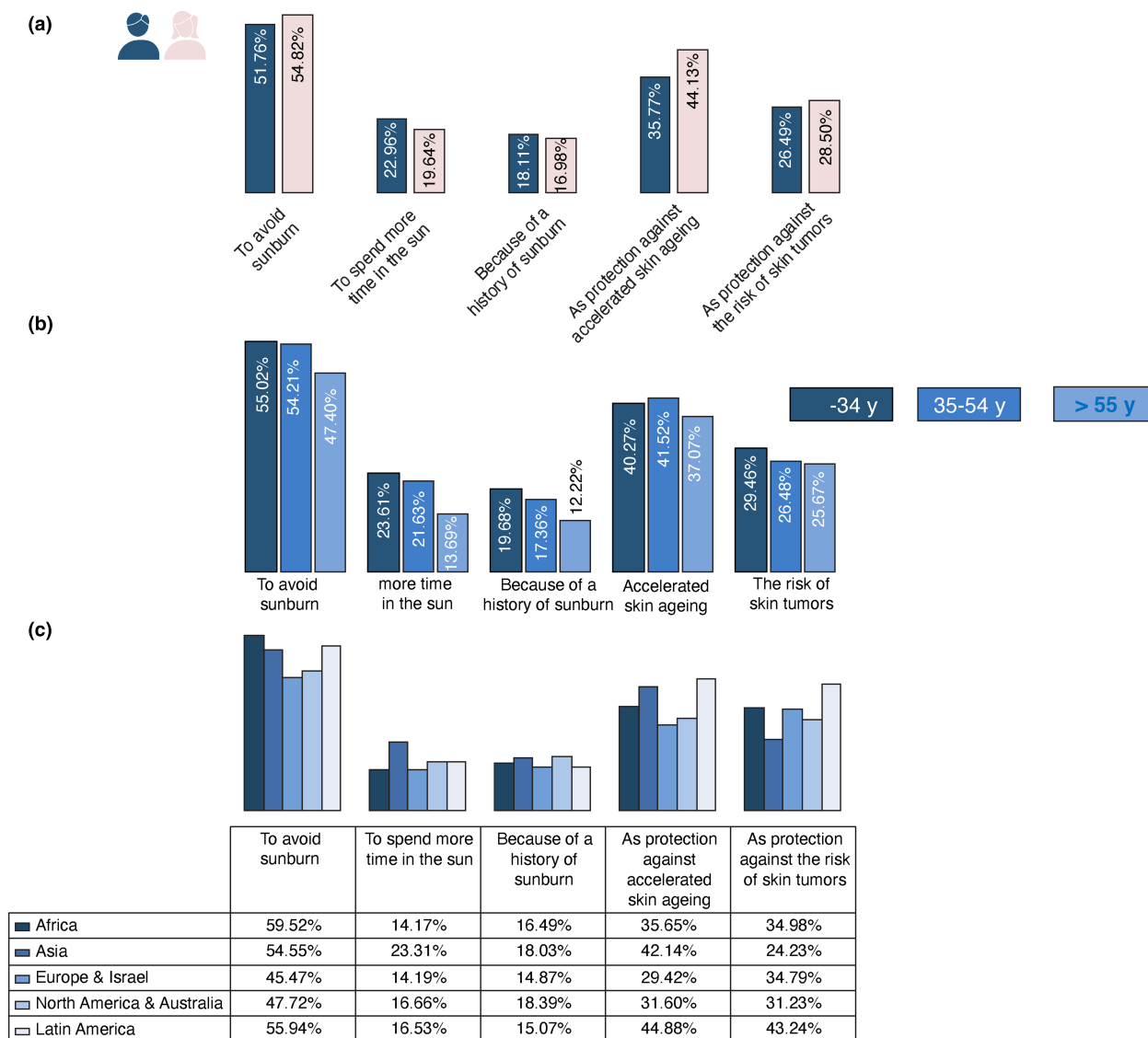


FIGURE 6 Underlying reasons contributing to sun protection according to (a) sex, (b) age and (c) region.

advertising and customization, allowing campaigns to tailor messages to specific demographics, locations or interests, with the possibility of sponsoring content during specific hours (between 10 AM and 4 PM for example).

Another important aspect to consider is the impact of pricing, as it has a significant influence on sunscreen application. Roughly one in five individuals reported not applying sunscreen due to cost, particularly in Africa where cost was three times more likely to be reported as a reason for not applying sunscreen compared to Europe and North America. The overall economic conditions and average income levels in a particular country or region can influence the affordability of sunscreen. While Africa is a diverse continent with significant variations in economic development, Africa has a higher prevalence of poverty compared to other regions. Poverty affects individuals' purchasing power and their ability to afford various goods and services, including sunscreen. While studies performed in Europe and North America conclude that

sunscreens are generally affordable,^{23,24} it is worth noting that there are national initiatives to provide more affordable or even free sunscreens.^{25,26} Efforts should also be geared towards increasing access to affordable sunscreen in regions with higher sun exposure, including Africa and Latin America.

Our study results are consistent with prior reports indicating that males are less likely to adopt sun-protective behaviours.^{27,28} Sun protection messaging frequently often targets females and emphasizes beauty standards related to ageing and sun damage.^{27,29,30} For instance, magazines targeting predominantly female audiences tend to feature advertisements for sun protection products, whereas magazines targeting predominantly male audiences have significantly fewer or no such advertisements.³¹ This is reflected in our results whereby 44% of females compared to 36% of males report using sunscreen as a protection against accelerated skin ageing. The formulation of sunscreen can also have a negative impact on its use among

males. The oily consistency of sunscreen and skin irritation have been reported to be main reasons deterring them from using sunscreen.³²

Interestingly, protection against accelerated skin ageing was a better incentive to using sunscreen than prevention from skin cancer. This was particularly true in females (44% vs. 28%, respectively) and in participants from Asia (42% vs. 24%, respectively). It has been suggested that individuals may perceive themselves as more susceptible to developing wrinkles and age spots compared to cancer, potentially due to the higher prevalence and visibility of the former.³³ In one study performed on 35 female British students, aged 18–34, with the majority having a skin type of II, III or IV, viewing morphed UV-aged images of themselves motivated them to change either their sun protection and/or their UV exposure behaviours.³⁴ These findings imply that appearance-based interventions can play a substantial role in healthcare and educational settings concerning sun protection behaviour. It is also essential to highlight that the immediate effects of sun exposure can have a great impact on sun protective behaviour, with more than 50% reporting using sunscreen to avoid sunburn (up to 59% in Africa and 56% in Latin America).

One in five participants reported applying sunscreen to spend more time in the sun. This finding aligns with previous reports. Indeed, it has been demonstrated that sunscreen use leads to longer duration of sun exposure when sun exposure is intentional, but not when sun exposure is not intentional.³⁵ Sunburns tend to be paradoxically more frequent among sunscreen users.

The strength of our study was the large sample from an international representative sample across five continents. Nevertheless, Internet-based questionnaires are a well-accepted and reliable method for collecting quantitative data, and allows wider access to geographically distant participants, as achieved in our study. Another limitation is that the questionnaire did not ask about profession, particularly about outdoor professions such as construction workers, farmers or lifeguards who are more likely to be exposed to the sun throughout the day. Although the studied population was representative of the population in each country based on socio-economic-cultural level, academic, and climate levels, we did not perform a comparison based on these variables. Further studies in each country may help tailor specific recommendations to each country or region. For instance, in a large part of Africa, people do not have any access to the internet, often not even electricity, nor is there sunscreen on sale, in addition to the fact that sun exposure is not a problem, as the largest portion of the population contains sufficient melanin for natural protection, and in some other countries, the cult of tanning is part of local culture, for example in Brazil. Lastly, our study is limited by recall bias, particularly concerning the 12-month timeframe, and it was carried out in winter/early spring in the northern hemisphere and summer/early autumn in the southern hemisphere. The study does not.

CONCLUSIONS

People across the world fail to consistently avoid sun exposure and use sunscreen, despite recognizing the significance and potential repercussions of not abiding by sun protective measures. Forgetfulness and the challenges posed by time constraints seem to be the biggest barriers to proper adherence. This should prompt health professionals and health authorities to tailor their message in a way that resonates with those who require persuasion, ensuring it is effectively communicated to the target audience. Pricing sunscreens reasonably and developing formulations to reduce the inconvenience of their application are challenges that need to be addressed by the manufacturers.

AUTHOR CONTRIBUTIONS

Charles Taieb: Conceptualization, data curation, formal analysis, funding acquisition; investigation; methodology; project administration; supervision; writing – review and editing. Yaron Ben Hayoun: Formal analysis, project administration. Henry W. Lim, Marketa Saint Aroman, Charbel Skayen, Marie Aleth Richard, Bruno Halioua, Nuria Perez Cullell, Catherine Baissac Khaled Ezzedine: Conceptualization, methodology; writing – review and editing. Christina Bergqvist: Writing – original draft.

ACKNOWLEDGEMENTS

None.

FUNDING INFORMATION

This study was granted by the Patient Centricity Pierre Fabre.

CONFLICT OF INTEREST STATEMENT

Henry W. Lim has no conflicts of interest to declare in this study. Marketa Saint Aroman is employed by the Pierre Fabre Laboratory. Charbel Skayen has no conflicts of interest to declare in this study. Bruno Halioua has no conflicts of interest to declare in this study. Nuria Perez Cullell is employed by the Pierre Fabre Laboratory. Yaron Ben Hayoun has no conflicts of interest to declare in this study. Catherine Baissac is employed by the Pierre Fabre Laboratory. Christina Bergqvist has no conflicts of interest to declare in this study. Charles Taieb has no conflicts of interest to declare in this study. Marie-Aleth Richard has no conflicts of interest to declare in this study. Khaled Ezzedine has no conflicts of interest to declare in this study.

DATA AVAILABILITY STATEMENT

The data sets generated during and/or analysed during this study are available from the authors upon reasonable request.

ORCID

Marketa Saint Aroman  <https://orcid.org/0000-0001-6124-5122>

Charbel Skayem  <https://orcid.org/0000-0002-4498-0683>

Bruno Halioua  <https://orcid.org/0000-0002-0823-6750>

Christina Bergqvist  <https://orcid.org/0000-0003-4985-6785>
 Charles Taieb  <https://orcid.org/0000-0002-5142-2479>
 Marie-Aleth Richard  <https://orcid.org/0000-0002-0870-9132>
 Khaled Ezzedine  <https://orcid.org/0000-0002-5468-4589>

REFERENCES

- Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, Jemal A, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2021;71(3):209–49. <https://doi.org/10.3322/caac.21660>
- Urban K, Mehrmal S, Uppal P, Giesey RL, Delost GR. The global burden of skin cancer: a longitudinal analysis from the Global Burden of Disease Study, 1990–2017. *JAAD Int*. 2021;2:98–108. <https://doi.org/10.1016/j.jdin.2020.10.013>
- Lomas A, Leonardi-Bee J, Bath-Hextall F. A systematic review of worldwide incidence of nonmelanoma skin cancer. *Br J Dermatol*. 2012;166(5):1069–80. <https://doi.org/10.1111/j.1365-2133.2012.10830.x>
- World Health Organization. Radiation: ultraviolet (UV) radiation and skin cancer. World Health Organization; n.d. Available from: [https://www.who.int/news-room/questions-and-answers/item/radiation-ultraviolet-\(uv\)-radiation-and-skin-cancer#:~:text=Currently%2C%20between%20%20and%203,skin%20cancer%20in%20their%20lifetime](https://www.who.int/news-room/questions-and-answers/item/radiation-ultraviolet-(uv)-radiation-and-skin-cancer#:~:text=Currently%2C%20between%20%20and%203,skin%20cancer%20in%20their%20lifetime). Accessed 01 Nov 2023.
- Arnold M, de Vries E, Whiteman DC, Jemal A, Bray F, Parkin DM, et al. Global burden of cutaneous melanoma attributable to ultraviolet radiation in 2012. *Int J Cancer*. 2018;143(6):1305–14. <https://doi.org/10.1002/ijc.31527>
- Islami F, Goding Sauer A, Miller KD, Siegel RL, Fedewa SA, Jacobs EJ, et al. Proportion and number of cancer cases and deaths attributable to potentially modifiable risk factors in the United States. *CA Cancer J Clin*. 2018;68(1):31–54. <https://doi.org/10.3322/caac.21440>
- Arnold M, Kvaskoff M, Thuret A, Guénel P, Bray F, Soerjomataram I. Cutaneous melanoma in France in 2015 attributable to solar ultraviolet radiation and the use of sunbeds. *J Eur Acad Dermatol Venereol*. 2018;32(10):1681–6. <https://doi.org/10.1111/jdv.15022>
- Keim U, Gandini S, Amaral T, Katalinic A, Holleczer B, Flatz L, et al. Cutaneous melanoma attributable to UVR exposure in Denmark and Germany. *Eur J Cancer*. 2021;159:98–104. <https://doi.org/10.1016/j.ejca.2021.09.0444>
- US Department of Health and Human Services. The surgeon general's call to action to prevent skin cancer. Washington, DC: Office of the Surgeon General (US); 2014. [cited 2022 July 7]. Available from: <http://www.ncbi.nlm.nih.gov/books/NBK247172/>
- Asai Y, Armstrong D, McPhie ML, Xue C, Rosen CF. Systematic review of interventions to increase awareness of ultraviolet radiation-induced harm and protective behaviors in post-secondary school adults. *J Cutan Med Surg*. 2021;25(4):424–36. <https://doi.org/10.1177/1203475420988863>
- McNoe BM, Morgaine KC, Reeder AI. Effectiveness of sun protection interventions delivered to adolescents in a secondary school setting: a systematic review. *J Skin Cancer*. 2021;2021:6625761. <https://doi.org/10.1155/2021/6625761>
- Niu Z, Bhurosy T, Heckman CJ. Digital interventions for promoting sun protection and skin self-examination behaviors: a systematic review. *Prev Med Rep*. 2022;26:101709. <https://doi.org/10.1016/j.pmedr.2022.101709>
- Oyebanjo E, Bushell F. A critical evaluation of the UK SunSmart campaign and its relevance to black and minority ethnic communities. *Perspect Public Health*. 2014;134(3):144–9. <https://doi.org/10.1177/1757913913516288>
- Køster B, Meyer M, Andersson T, Engholm G, Dalum P. Development in sunburn 2007–2015 and skin cancer projections 2007–2040 of campaign results in the Danish population. *Medicine (Baltimore)*. 2018;97(41):e12738. <https://doi.org/10.1097/MD.00000000000012738>
- Misery L, Morisset S, Séité S, Brenaut E, Ficheux AS, Fluhr JW, et al. Relationship between sensitive skin and sleep disorders, fatigue, dust, sweating, food, tobacco consumption or female hormonal changes: results from a worldwide survey of 10 743 individuals. *J Eur Acad Dermatol Venereol*. 2021;35(6):1371–6. <https://doi.org/10.1111/jdv.17162>
- Misery L, Reaux-Le Goazigo A, Morisset S, Seite S, Delvigne V, Cochener B, et al. Association of sensitive eyes with sensitive skin: a worldwide study of 10,743 subjects. *Skin Pharmacol Physiol*. 2022;35(3):148–55. <https://doi.org/10.1159/000522056>
- Green AC, Williams GM, Logan V, Strutton GM. Reduced melanoma after regular sunscreen use: randomized trial follow-up. *J Clin Oncol*. 2011;29(3):257–63. <https://doi.org/10.1200/JCO.2010.28.7078>
- Waldman RA, Grant-Kels JM. The role of sunscreen in the prevention of cutaneous melanoma and nonmelanoma skin cancer. *J Am Acad Dermatol*. 2019;80(2):574–6.e1. <https://doi.org/10.1016/j.jaad.2018.06.069>
- Perez M, Abisaad JA, Rojas KD, Marchetti MA, Jaimes N. Skin cancer: primary, secondary, and tertiary prevention. Part I. *J Am Acad Dermatol*. 2022;87(2):255–68. <https://doi.org/10.1016/j.jaad.2021.12.066>
- Armstrong AW, Watson AJ, Makredes M, Frangos JE, Kimball AB, Kvedar JC. Text-message reminders to improve sunscreen use: a randomized, controlled trial using electronic monitoring. *Arch Dermatol*. 2009;145(11):1230–6. <https://doi.org/10.1001/archdermatol.2009.269>
- Chamberg-Michilot D, Tellez WA, Becerra-Chauca N, Zafra-Tanaka JH, Taype-Rondan A. Text message reminders for improving sun protection habits: a systematic review. *PLoS One*. 2020;15(5):e0233220. <https://doi.org/10.1371/journal.pone.0233220>
- Vasconcelos Silva C, Jayasinghe D, Janda M. What can twitter tell us about skin cancer communication and prevention on social media? *Dermatology*. 2020;236(2):81–9. <https://doi.org/10.1159/000506458>
- Mahé E, Beauchet A, de Maleissye MF, Saiag P. Are sunscreens luxury products? *J Am Acad Dermatol*. 2011;65(3):e73–e79. <https://doi.org/10.1016/j.jaad.2010.06.043>
- Johal R, Leo MS, Ma B, Sivamani RK. The economic burden of sunscreen usage. *Dermatol Online J*. 2014;20(6):13030/qt6v0352fw.
- Available from: <https://www.theguardian.com/world/2023/jun/12/netherlands-free-sun-cream-record-skin-cancer-levels>. Accessed 01 Nov 2023.
- Available from: <https://www.theguardian.com/society/2023/may/15/charities-call-for-scrapping-of-vat-on-sunscreen-amid-skin-cancer-fears>. Accessed 01 Nov 2023.
- Abroms L, Jorgensen CM, Southwell BG, Geller AC, Emmons KM. Gender differences in young adults' beliefs about sunscreen use. *Health Educ Behav*. 2003;30(1):29–43. <https://doi.org/10.1177/1090198102239257>
- McKenzie C, Nahm WJ, Kearney CA, Zampella JG. Sun-protective behaviors and sunburn among US adults. *Arch Dermatol Res*. 2023;315(6):1665–74. <https://doi.org/10.1007/s00403-023-02547-z>
- Jackson KM, Aiken LS. Evaluation of a multicomponent appearance-based sun-protective intervention for young women: uncovering the mechanisms of program efficacy. *Health Psychol*. 2006;25(1):34–46. <https://doi.org/10.1037/0278-6133.25.1.34>
- Matts PJ, Fink B. Chronic sun damage and the perception of age, health and attractiveness. *Photochem Photobiol Sci*. 2010;9(4):421–31. <https://doi.org/10.1039/b9pp00166b>
- Wiznia LE, Wang J, Steuer AB, Elbuluk N. Deficiency of sun protection advertising exists in consumer magazines across demographic groups and varies by target demographic. *J Am Acad Dermatol*. 2019;80(4):1139–41. <https://doi.org/10.1016/j.jaad.2018.10.004>
- Roberts CA, Goldstein EK, Goldstein BG, Jarman KL, Paci K, Goldstein AO. Men's attitudes and behaviors about skincare and sunscreen use behaviors. *J Drugs Dermatol*. 2021;20(1):88–93. <https://doi.org/10.36849/JDD.5470>

33. Hevey D, Pertl M, Thomas K, Maher L, Craig A, Chuinneagain SN. Body consciousness moderates the effect of message framing on intentions to use sunscreen. *J Health Psychol.* 2010;15(4):553–9. <https://doi.org/10.1177/1359105309355335>
34. Williams AL, Grogan S, Buckley E, Clark-Carter D. A qualitative study examining women's experiences of an appearance-focussed facial-ageing sun protection intervention. *Body Image.* 2012;9(3):417–20. <https://doi.org/10.1016/j.bodyim.2012.04.008>
35. Autier P, Boniol M, Doré JF. Sunscreen use and increased duration of intentional sun exposure: still a burning issue. *Int J Cancer.* 2007;121(1):1–5. <https://doi.org/10.1002/ijc.22745>

How to cite this article: Lim HW, Saint Aroman M, Skayem C, Halioua B, Perez Cullell N, Ben Hayoun Y, et al. Sun exposure and protection habits: Self-reported attitudes, knowledge and behaviours. *J Eur Acad Dermatol Venereol.* 2024;38:2024–2033. <https://doi.org/10.1111/jdv.20080>

APUNTES

INTRO ARTÍCULO DUCCIÓN:

- El cáncer de piel es una preocupación mundial creciente debido a que la exposición a la radiación UV es el principal factor de riesgo
- OMS: el aumento de las actividades al aire libre y de la exposición recreativa representa un riesgo significativo para la salud.
- Los melanomas se encuentran entre los tipos de cáncer con la mayor proporción de casos prevenibles. OMS ha proporcionado directrices para la protección solar, como usar protectores solares de amplio espectro cada dos horas al aire libre, etc.
- Objetivo: determinar e informar sobre la prevalencia de actitudes, conocimientos y comportamientos autoinformados con respecto a dos de las principales recomendaciones de protección solar (evitar la exposición al sol y usar protectores solares) en una muestra representativa internacional de cinco continentes.

METODOLOGÍA:

- Población de estudio: estudio transversal multinacional (se limita a observar y registrar las características o prevalencias encontradas, facilitando la comparación de frecuencias de un evento entre distintas naciones).
- Instrumento: Encuesta en línea realizada por la empresa de sondeos HC Conseil Paris (Francia) entre enero y abril de 2023 en 20 países.
- Observación del instrumento: Sin aprobación de un comité de ética institucional, ya que no implicó ningún examen clínico y utilizó datos anonimizados.
- La selección de participantes tuvo como objetivo asegurar la representatividad nacional considerando factores como la edad (mayor de 16 años), el género, la región administrativa, el entorno (incluidas las grandes ciudades, los pueblos y las zonas rurales) y la distribución de ingresos en cada país, como se describió previamente en estudios anteriores.
- La población de estudio es de 900 000 usuarios de Internet mayores de 16 años que estaban dispuestos a participar, de los cuales se seleccionó una muestra aleatoria de 50 552 personas.
- Los 20 países se agruparon en cinco regiones: América del Norte y Australia (Australia, Canadá y Estados Unidos), América Latina (Brasil y México), Europa (Dinamarca, Francia, Alemania, Italia, Israel, Polonia, Portugal y España), África (Kenia, Senegal, Sudáfrica y Emiratos Árabes Unidos) y Asia (China, India y Corea del Sur).
- Variables cualitativas representadas como números (porcentajes). Variable cualitativa con mediana o media desviación estándar.
- Grupación, Categorías de edades (3), Conductas de protección solar y características demográficas
- Pruebas aplicadas: Chi cuadrado (the chi-squared test) o la prueba exacta de Fisher (the Fisher's exact test), según correspondiera
- Los análisis fueron realizados con RStudio (R4.1.3).

RESULTADOS:

El estudio analizó a 50552 participantes de 20 países en 5 continentes (25388 hombres y 25164 mujeres). Exposición voluntaria al sol: Un 83,2% reportó haberse expuesto voluntariamente al sol (bronceado/recreación) al menos una vez en los últimos 12 meses.

Mayor frecuencia en hombres (85.04%) en comparación con las mujeres (82.38%) ($p < 0.001$). Por regiones: África presentó la prevalencia más alta (90.97%), seguida de Latinoamérica (88.07%), Asia (85.03%), Norteamérica/Australia (78.02%) y Europa (71.62%).

Por edad: La exposición voluntaria fue mayor en el grupo de 16 a 34 años (87.51%) y disminuyó progresivamente en los grupos de 35 a 54 años (82.41%) y mayores de 55 años (77.51%) ($p < 0.001$).

Exposición en horas de alta radiación (10 AM a 4 PM): El 47.96% de la población admitió exponerse en este rango horario crítico. La prevalencia fue superior en hombres (49.33%) que en mujeres (46.53%) ($p < 0.001$) y más acentuada en jóvenes de 16 a 34 años (53.75%). La principal causa reportada para no evitar este horario fue la conveniencia de la hora (32.28%). Uso y reaplicación de fotoprotector: Solo el 24.05% de los participantes reportó aplicarse protector solar cada 2 horas cuando se encuentra al aire libre. El motivo principal para el incumplimiento de la reaplicación fue el olvido (27.79%). Los hombres y las personas de mayor edad mostraron una probabilidad significativamente menor de adoptar conductas de fotoprotección adecuadas.

DISCUSIÓN:

A pesar del desarrollo masivo de campañas de concientización pública sobre los riesgos de la radiación ultravioleta y el cáncer de piel, su impacto real en la reducción de las tasas de incidencia y mortalidad por melanoma y carcinoma sigue siendo limitado. Falla en la traducción del conocimiento a la práctica: El estudio demuestra que la falta de cumplimiento de las recomendaciones sanitarias (evitar horas pico y reaplicar fotoprotector) no responde principalmente a la ignorancia de los riesgos, sino a barreras conductuales y logísticas del día a día. Patrones de riesgo demográfico: Los hombres y los grupos más jóvenes exhiben las tasas más altas de exposición solar voluntaria e inadvertida en horas críticas, mientras que los hombres en general y los adultos mayores muestran menor disposición a integrar hábitos estrictos de fotoprotección. Naturaleza de las barreras: Las dos principales limitantes identificadas —la conveniencia horaria para la exposición y el olvido para la reaplicación— revelan que las recomendaciones actuales resultan poco prácticas o difíciles de mantener dentro de las rutinas cotidianas de la población.

CONCLUSIONES:

Necesidad de replantear las estrategias de salud pública: Las campañas educativas basadas únicamente en transmitir información sobre los peligros de la radiación UV han alcanzado un techo de efectividad, por lo que es necesario diseñar intervenciones dirigidas a superar las barreras pragmáticas específicas. Alianza estratégica entre salud e industria: Para mejorar la adherencia de la población, los autores concluyen que debe existir una colaboración directa entre las autoridades sanitarias y la industria dermocosmética. Mejora en la accesibilidad y formulación: Las soluciones requieren establecer precios más accesibles para los fotoprotectores y desarrollar formulaciones cosméticas que disminuyan la incomodidad o la carga asociada a la aplicación frecuente